Database Resiliency

Ensuring we back up our production database regularly is critical to ensure we can recover in case of error; it's possible to back up a database by running the normal database dump command inside a container. However, what's slightly trickier is how to make this happen automatically in production.

There are a number of different ways to handle this:

- *Platform-specific.* Some container platforms allow you to schedule containers (for example, Amazon ECS scheduled tasks).[30] Using these schedulers, you can run containers to back up the database at regular intervals. Additionally, platforms may offer backup capabilities; for example, Amazon Elastic Block Store (Amazon EBS) volumes provide automated incremental snapshotting capabilities.[31] This can be a low-hassle, reliable approach to maintaining backups.
- *Cron running on the Dockerhost.* There's nothing stopping you from setting up `cron` or a similar scheduler on your Dockerhost that triggers a container (or noncontainerized script) for backing up the database. Some people like this approach, particularly because of its simplicity. However, the downside is the risk that your Dockerhost becomes a special snowflake that's harder to maintain. Your database backup mechanism is living outside of your containerization, so you lose all the benefits that brings.
- *Use third-party tools.* For example, Barman for Postgres.[32]

Containers on Autopilot

There's a broader approach that's beginning to emerge known as *the autopilot pattern*.[33] This involves baking standard operational tasks (such as scaling and resiliency) directly into your containerized services themselves.

Rather than maintaining this operational logic spread externally with schedulers and separate task-based containers, your app containers have the smarts to perform their own life-cycle management. For example, imagine launching a Postgres container configured to check if its database was populated, and if it finds that it wasn't, goes and fetches and restores the latest backup. Done well, maintenance and resiliency become automatic.

30. https://docs.aws.amazon.com/AmazonECS/latest/developerguide/scheduled_tasks.html
31. https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/snapshot-lifecycle.html
32. https://www.pgbarman.org/about/
33. http://autopilotpattern.io